

Intro to ML - VisionGATE Questions

GATE Data Science and Artificial Intelligence (GATE DA)

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1 Task Taxonomy

- Q1.** In the context of machine learning, what is a *model*?
- (A) A dataset used to train an algorithm
 - (B) A physical device that makes predictions
 - (C) A mathematical representation of reality
 - (D) The set of labels associated with the data
- Q2.** Which of the following is a regression problem?
- (A) Identifying whether an email is spam
 - (B) Classifying images as cat or dog
 - (C) Predicting whether a customer will churn
 - (D) Predicting the price of a house from its area
- Q3.** Which of the following is an example of a machine learning problem?
- (A) Computing the average of a list of numbers
 - (B) Sorting records alphabetically
 - (C) Finding the shortest path using Dijkstra's algorithm
 - (D) Detecting whether an incoming email is spam
- Q4.** What kind of values do the parameters of a regression model take?
- (A) Binary values only
 - (B) Non-negative integers only
 - (C) Real values
 - (D) Values restricted to $[0, 1]$
- Q5.** Which of the following are **classification** problems? (Select all that apply.)
- (A) Predicting whether a tumour is benign or malignant
 - (B) Predicting tomorrow's rainfall in millimetres
 - (C) Predicting which of five categories a news article belongs to
 - (D) Predicting the resale value of a used car
- Q6.** Which of the following is a regression problem?
- (A) Predicting the temperature tomorrow in degrees Celsius
 - (B) Predicting whether a transaction is fraudulent
 - (C) Predicting the language a document is written in
 - (D) Predicting whether a student passes or fails

Q7. Which of the following are **unsupervised** learning problems? (Select all that apply.)

- (A) Grouping customers into segments based on purchase history
- (B) Compressing high-dimensional images into a smaller representation
- (C) Predicting house prices from labelled past sales
- (D) Classifying handwritten digits using a labelled dataset

Q8. A model outputs $\hat{y} \in \{-1, +1\}$ for each input. What kind of problem is this?

- (A) Regression
- (B) Clustering
- (C) Dimensionality reduction
- (D) Binary classification

Q9. A model predicts a student's grade as one of A, B, C, D, or F. What kind of problem is this?

- (A) Regression
- (B) Classification
- (C) Clustering
- (D) Dimensionality reduction

Q10. Consider two problems.

Problem 1: Predict whether a loan applicant will default (yes/no).

Problem 2: Predict the amount of the loan a bank should sanction, in rupees. What is the key difference between them?

- (A) Problem 1 is unsupervised; Problem 2 is supervised
- (B) Both are regression problems, differing only in scale
- (C) In Problem 1 the target is discrete; in Problem 2 it is continuous
- (D) Problem 1 requires labelled data; Problem 2 does not

2 Norms

Q1. If $\|\mathbf{v}\| = 5$ and $\|\mathbf{w}\| = 3$, what are the smallest and largest possible values of $\|\mathbf{v} - \mathbf{w}\|$ and of $\mathbf{v} \cdot \mathbf{w}$?

- (A) $0 \leq \|\mathbf{v} - \mathbf{w}\| \leq 8$ and $0 \leq \mathbf{v} \cdot \mathbf{w} \leq 15$
- (B) $2 \leq \|\mathbf{v} - \mathbf{w}\| \leq 8$ and $-15 \leq \mathbf{v} \cdot \mathbf{w} \leq 15$
- (C) $2 \leq \|\mathbf{v} - \mathbf{w}\| \leq 8$ and $0 \leq \mathbf{v} \cdot \mathbf{w} \leq 15$
- (D) $3 \leq \|\mathbf{v} - \mathbf{w}\| \leq 5$ and $-15 \leq \mathbf{v} \cdot \mathbf{w} \leq 15$

Q2. Let $\mathbf{x} = [3, -4, 12, 0]^\top$. What are $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$ and $\|\mathbf{x}\|_\infty$ respectively?

- (A) 19, 13, 12
- (B) 19, 169, 12
- (C) 11, 13, 12
- (D) 19, 13, 3

Q3. Let $\mathbf{x} = [-1, 2, -2, 4, -4]^\top$. Which of the following is the **greatest**?

- (A) The L_1 norm
- (B) The L_2 norm
- (C) The L_3 norm
- (D) The L_∞ norm

Q4. For $\mathbf{x} = [1, 1, 1, 1]^\top$, what is the value of the ratio $\frac{\|\mathbf{x}\|_1}{\|\mathbf{x}\|_2}$?

- (A) 1
- (B) 2
- (C) 4
- (D) $\frac{1}{2}$

Q5. For which of the following vectors in $\mathbb{R}^4$ is the ratio $\frac{\|\mathbf{x}\|_1}{\|\mathbf{x}\|_2}$ the **smallest**?

- (A) $[1, 1, 1, 1]^\top$
- (B) $[3, 0, 0, 0]^\top$
- (C) $[2, 2, 1, 0]^\top$
- (D) $[1, 1, 1, 0]^\top$

Q6. Let $\mathbf{u} = [2, -3, 6]^\top$ and $\mathbf{v} = [5, 1, 2]^\top$. What are the L_1 , L_2 and L_∞ distances between $\mathbf{u}$ and $\mathbf{v}$ respectively? (Round to 4 decimal places where needed.)

- (A) 11, 6.4031, 4
- (B) 11, 41, 4
- (C) 17, 10.8167, 8
- (D) 4, 6.4031, 11

Q7. Which of the following is true for **every** non-zero vector $\mathbf{x} \in \mathbb{R}^d$?

- (A) $\|\mathbf{x}\|_1 \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_\infty$
- (B) $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$
- (C) $\|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_1$
- (D) The ordering depends on the values of $\mathbf{x}$

Q8. For a non-zero vector $x \in \mathbb{R}^d$, what is the range of the ratio $\frac{\|x\|_1}{\|x\|_2}$? Here, $\|x\|_1$ is the L1-norm and $\|x\|_2$ is the L2-norm of vector x .

- (A) Between 1 and d
- (B) Between 1 and $\sqrt{d}$
- (C) Between $\sqrt{d}$ and d
- (D) Between $\frac{1}{\sqrt{d}}$ and $\sqrt{d}$

Q9. Define $f(x)$ = the number of non-zero entries of x (the “L0 norm”). Exactly one norm axiom fails. Which one?

- (A) Absolute homogeneity
- (B) The triangle inequality
- (C) Definiteness: there exist non-zero vectors with $f(x) = 0$
- (D) Both definiteness and the triangle inequality

Q10. Which of the following functions on $\mathbb{R}^d$ are valid norms? (Choose all the correct answers.)

- (A) $f(\mathbf{x}) = \max_i |x_i|$
- (B) $f(\mathbf{x}) = \text{the number of non-zero entries of } \mathbf{x}$
- (C) $f(\mathbf{x}) = \left(\sum_i |x_i|^{1/2}\right)^2$
- (D) $f(\mathbf{x}) = \sum_i |x_i|$

3 Vectorization and Dimensions

Q1. A colour (RGB) image of size 512×512 is fed to an algorithm that classifies it into one of 10 categories. The output is a vector of 10 class scores. Based on the above scenario, answer the following:

- (i) If the input image is flattened into a vector $\mathbf{x}$, what is its length?
 - (A) 262144
 - (B) 786432
 - (C) 1536
 - (D) 512
- (ii) If the output is a vector $\mathbf{y}$ of class scores, what is its length?
 - (A) 1
 - (B) 3
 - (C) 10
 - (D) 30
- (iii) The model is $\mathbf{y} = A\mathbf{x} + \mathbf{b}$. How many elements does the matrix A contain?
 - (A) 786442
 - (B) 2621440
 - (C) 7864320
 - (D) 7864330
- (iv) How many learnable parameters does the model have in total, counting both A and $\mathbf{b}$?
 - (A) 7864320
 - (B) 7864330
 - (C) 7864322
 - (D) 786442
- (v) The model is trained on a batch of 500 such images. If the design matrix X stores one image per row, how many entries does X contain in total?
 - (A) 393216000
 - (B) 1572864
 - (C) 786932
 - (D) 131072000

Q2. A video clip consists of 60 frames, each a colour (RGB) image of size 128×128 . Based on the above scenario, answer the following:

- (i) Flattening the entire clip into a single vector, what is its length?

- (A) 983040
(B) 2949120
(C) 49152
(D) 16384
- (ii) A model compresses this clip to a 256-dimensional representation using $\mathbf{z} = A\mathbf{x}$. What is the shape of A , written as (rows $\times$ columns)?
- (A) 2949120×256
(B) 256×2949120
(C) 256×49152
(D) 60×256
- (iii) A grayscale image of size 128×128 is flattened into $\mathbf{x}$, and a model computes $\mathbf{y} = A\mathbf{x} + \mathbf{b}$ where $\mathbf{y} \in \mathbb{R}^{64}$. Which of the following is correct?
- (A) A is 16384×64 and $\mathbf{b} \in \mathbb{R}^{16384}$
(B) A is 64×16384 and $\mathbf{b} \in \mathbb{R}^{64}$
(C) A is 128×128 and $\mathbf{b} \in \mathbb{R}^{64}$
(D) A is 64×128 and $\mathbf{b} \in \mathbb{R}^{128}$

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4 Statistics of a Dataset

Q1. A dataset has 4 samples and 3 features, given below:

Sample	X_1	X_2	X_3
1	2	4	1
2	4	6	3
3	6	10	2
4	8	12	6

Use the **sample** convention (divide by $n - 1$) throughout, and round to 4 decimal places where needed. Based on the above dataset, answer the following:

- (i) What is the mean of feature X_2 ?
 - (A) 6
 - (B) 8
 - (C) 10
 - (D) 10.6667
- (ii) What is the sample variance of feature X_1 ?
 - (A) 5
 - (B) 6.6667
 - (C) 20
 - (D) 2.5820
- (iii) What is the sample covariance between X_1 and X_2 ?
 - (A) 7
 - (B) 9.3333
 - (C) 28
 - (D) 13.3333
- (iv) What is the correlation coefficient between X_1 and X_2 ?
 - (A) 0.9899
 - (B) 0.8367
 - (C) 0.7606
 - (D) 1.0000
- (v) What are the dimensions of the covariance matrix Σ for this dataset?
 - (A) 4×4
 - (B) 3×3
 - (C) 4×3
 - (D) 3×4
- (vi) What is $\text{Var}[X_1 + X_2]$?

- (A) 20.0000
(B) 29.3333
(C) 38.6667
(D) 1.3333
- (vii) What is the trace of Σ ?
(A) 24.6667
(B) 20.0000
(C) 4.6667
(D) 13.3333
- Q2.** Consider a dataset with n samples and d features, with covariance matrix Σ . Answer the following:
- (i) Suppose every value of X_1 is multiplied by 100. Which of the following is correct?
(A) $\text{Var}[X_1]$ is multiplied by 100; the correlation ρ_{12} is unchanged
(B) $\text{Var}[X_1]$ is multiplied by 10000; the correlation ρ_{12} is unchanged
(C) $\text{Var}[X_1]$ is multiplied by 10000; the correlation ρ_{12} is multiplied by 100
(D) Both the variance and the correlation are multiplied by 100
- (ii) Two features X and Y satisfy $\text{Cov}(X, Y) = 0$. Which of the following can be concluded?
(A) X and Y are independent
(B) X and Y have no linear relationship, but may still be dependent
(C) X and Y are identically distributed
(D) At least one of X and Y has zero variance
- (iii) Given $\text{Var}[X] = 9$, $\text{Var}[Y] = 4$ and $\text{Cov}(X, Y) = 4.5$, what are $\text{Var}[X - Y]$ and $\text{Var}[Y - X]$ respectively?
(A) 4 and 22
(B) 22 and 4
(C) 4 and 4
(D) 13 and 13
- (iv) Which of the following identities holds for **any** two random variables X and Y , with no assumption of independence?
(A) $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$
(B) $E[X + Y] = E[X] + E[Y]$
(C) $E[XY] = E[X]E[Y]$
(D) $\text{Var}[X - Y] = \text{Var}[X] - \text{Var}[Y]$